

PA4 - e-NFAs

(soluções)

1

Not all ϵ -NFAs can be systematically converted into a DFA
(1/1 Ponto)

- ☐ True
- ☒ False

2

To which state(s) does the following ϵ -NFA go at the very beginning? (i.e. $EClose(p)$) 
(0/1 Ponto)

	ϵ	0	1
$\rightarrow p$	$\{r\}$	$\{q\}$	$\{r\}$
q	$\emptyset$	$\{s, r\}$	$\{t\}$
r	$\{q\}$	$\{t\}$	$\{r\}$
s	$\emptyset$	$\{p\}$	$\emptyset$
*t	$\emptyset$	$\emptyset$	$\{t\}$

- ☒ p
- ☒ q
- ☒ r
- ☐ s
- ☐ t

3

How many acceptance states has the DFA that results from the conversion of the following ϵ -NFA? 

(1/1 Ponto)

	ϵ	0	1
$\rightarrow p$	$\{r\}$	$\{q\}$	$\{r\}$
q	$\emptyset$	$\{s, r\}$	$\{t\}$
r	$\{q\}$	$\{t\}$	$\{r\}$
s	$\emptyset$	$\{p\}$	$\emptyset$
*t	$\emptyset$	$\emptyset$	$\{t\}$

- ☐ 1
- ☒ 3
- ☐ 5
- ☐ 7

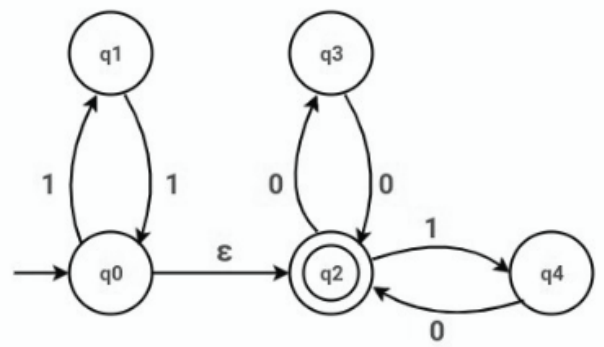
4

Considering an ϵ -NFA that creates multiple paths when processing a given input, how many of those paths must end in an acceptance state for the input to be accepted by the automaton?

(0/1 Ponto)

- ☐ All of them.
- ☐ Just one.
- ☒ At least one.
- ☐ It depends on the input and/or ϵ -NFA.

5

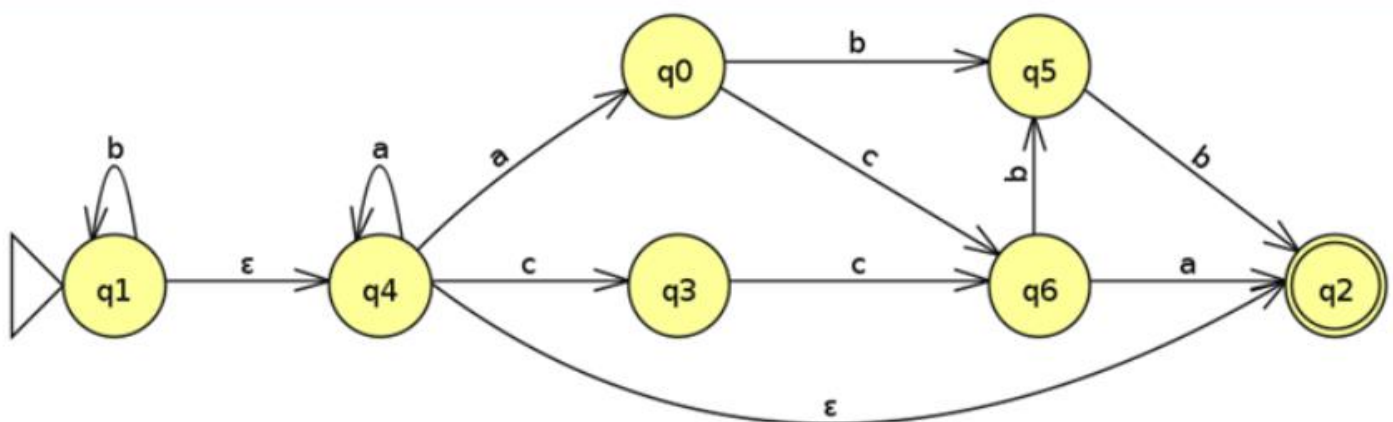


Which of the following inputs are accepted by the automaton?
(0/1 Ponto)

- ☒ 11
- ☐ 111
- ☐ 011
- ☒ 1110
- ☐ 0011
- ☒ 111010

6

Describe in english (or portuguese) the language specified by the following automaton.
(-/1 Ponto)



Write the formal notation for the automaton of the previous exercise (ex. 6). Present the transition function just for the states: q_1 , q_4 and q_0 .

(-/1 Ponto)